

BUS33-4

# Labour Economics & Inequalities

Wages, unemployment, and the distribution of income and wealth

---

The distributional turn: mechanisms plus the causal-inference standards the field requires.

Albert School — 22 hours

# Contents

|     |                                                                  |   |
|-----|------------------------------------------------------------------|---|
| 1   | Labour supply and demand                                         | 2 |
| 1.1 | Labour supply: the work–leisure trade-off                        | 2 |
| 1.2 | Labour demand: the marginal-productivity condition               | 3 |
| 1.3 | Capital–labour substitution                                      | 3 |
| 2   | Human capital                                                    | 3 |
| 2.1 | Becker’s model                                                   | 3 |
| 3   | Wage structure: differentials, discrimination, and decomposition | 4 |
| 3.1 | Compensating differentials (Rosen)                               | 4 |
| 3.2 | Statistical discrimination                                       | 4 |
| 3.3 | The Oaxaca–Blinder decomposition                                 | 4 |
| 4   | Unemployment: frictions, matching, and the natural rate          | 4 |
| 4.1 | Types of unemployment                                            | 4 |
| 4.2 | The Mortensen–Pissarides matching model                          | 5 |
| 4.3 | The Beveridge curve                                              | 5 |
| 4.4 | NAIRU                                                            | 5 |
| 5   | Institutional effects on equilibrium unemployment                | 5 |
| 5.1 | Minimum wage, unions, and unemployment insurance                 | 5 |
| 5.2 | The minimum-wage debate and the Card–Krueger reframing           | 6 |
| 6   | Inequality measurement                                           | 6 |
| 6.1 | The Lorenz curve                                                 | 6 |
| 6.2 | The Gini coefficient                                             | 6 |
| 7   | Wealth inequality and its dynamics                               | 7 |
| 7.1 | Top concentration and the $r > g$ mechanism                      | 7 |
| 7.2 | Intergenerational mobility                                       | 7 |
| 8   | Causal inference for labour-market policy                        | 7 |
| 8.1 | The counterfactual problem                                       | 7 |
| 8.2 | Natural experiments and instrumental variables                   | 7 |
| 8.3 | Difference-in-differences                                        | 7 |
| 9   | Automation and labour-market polarisation                        | 8 |
| 9.1 | Routine versus non-routine tasks                                 | 8 |
| 9.2 | Polarisation and complementarity                                 | 8 |

## 1 Labour supply and demand

### 1.1 Labour supply: the work–leisure trade-off

A worker allocates a fixed time endowment  $T$  between hours of work  $h$  and leisure  $\ell = T - h$ . With hourly wage  $w$  and non-labour income  $V$ , consumption is  $c = wh + V$  and the worker maximises utility  $u(c, \ell)$  subject to the time and budget constraints:

$$\max_{c, \ell} u(c, \ell) \quad \text{s.t.} \quad c = w(T - \ell) + V.$$

The optimum equates the marginal rate of substitution between leisure and consumption to the wage:

$$\frac{u_\ell}{u_c} = w.$$

The wage is the price (opportunity cost) of leisure.

**Definition 1 (Income and substitution effects of a wage change)** A rise in  $w$  has two opposing effects on hours worked:

- **Substitution effect:** leisure becomes more expensive relative to consumption, which raises desired hours.
- **Income effect:** a higher wage raises real income, and if leisure is a normal good this raises desired leisure, which lowers hours.

The net effect is ambiguous. When the income effect dominates at high wages the individual labour-supply curve bends backward.

## 1.2 Labour demand: the marginal-productivity condition

A competitive firm with production function  $Y = F(K, L)$  hires labour up to the point where the value of the marginal product of labour equals the wage. With output price  $p$  and marginal product  $MP_L = \partial F / \partial L$ :

$$p \cdot MP_L = w, \quad \text{equivalently} \quad MP_L = \frac{w}{p}.$$

The labour-demand curve is the downward-sloping marginal-product schedule: diminishing marginal product means more labour is demanded only at a lower real wage.

## 1.3 Capital–labour substitution

The elasticity of substitution  $\sigma$  measures how easily capital replaces labour as relative factor prices change. For inputs  $K$  and  $L$  with relative price  $w/r$ ,

$$\sigma = \frac{d \ln(K/L)}{d \ln(w/r)} \geq 0.$$

A large  $\sigma$  means a wage increase (or a labour-saving automation shock that lowers the effective price of capital) induces a large fall in labour demand; a small  $\sigma$  means labour and capital are complements and employment responds little.

# 2 Human capital

## 2.1 Becker's model

Becker treats schooling as an investment: a worker incurs costs today (direct costs plus forgone earnings) in exchange for higher future earnings. The investment is worthwhile when the present value of the additional earnings exceeds the present value of the costs.

**Definition 2 (Internal rate of return to education)** Let  $\Delta w_t$  be the extra earnings in year  $t$  attributable to an additional year of schooling,  $C$  the total cost (direct plus forgone earnings), and  $r$  the discount rate. The rate of return is the value  $r^*$  solving

$$C = \sum_{t=1}^N \frac{\Delta w_t}{(1 + r^*)^t}.$$

Education is worth undertaking when  $r^*$  exceeds the cost of funds.

On real earnings data the return is commonly estimated from the Mincer earnings equation, a log-linear regression of earnings on years of schooling  $s$  and experience:

$$\ln w = \beta_0 + \rho s + \beta_1 \exp + \beta_2 \exp^2 + \varepsilon,$$

where the coefficient  $\rho$  is read as the average rate of return to a year of schooling.

### 3 Wage structure: differentials, discrimination, and decomposition

#### 3.1 Compensating differentials (Rosen)

Compensating differentials are wage differences that offset the non-wage characteristics of jobs. Unpleasant, dangerous, or insecure jobs must pay more, all else equal, to attract workers; desirable job attributes are paid for through lower wages. Observed wage gaps may therefore reflect differences in job attributes rather than differences in treatment.

#### 3.2 Statistical discrimination

Statistical discrimination arises when employers, unable to observe a worker's individual productivity, use group averages as a proxy. Members of a group are then paid according to the group mean rather than their own productivity, which can sustain wage gaps even without taste-based prejudice.

#### 3.3 The Oaxaca–Blinder decomposition

The Oaxaca–Blinder method decomposes the mean wage gap between two groups (say  $A$  and  $B$ ) into a part explained by differences in observable characteristics and an unexplained residual.

**Definition 3 (Oaxaca–Blinder decomposition)** Estimate  $\ln w = X\beta + \varepsilon$  separately in each group, with mean characteristics  $\bar{X}_A, \bar{X}_B$  and coefficients  $\hat{\beta}_A, \hat{\beta}_B$ . The mean log-wage gap splits as

$$\overline{\ln w}_A - \overline{\ln w}_B = \underbrace{(\bar{X}_A - \bar{X}_B)\hat{\beta}_B}_{\text{explained}} + \underbrace{\bar{X}_A(\hat{\beta}_A - \hat{\beta}_B)}_{\text{unexplained}}.$$

The **explained** (endowments) component reflects differences in characteristics such as education and experience; the **unexplained** component reflects differences in returns to those characteristics.

**Common pitfall** The unexplained component is not a clean measure of discrimination. It also captures unobserved productivity differences and compensating differentials, and it is sensitive to which characteristics are included. A statistical-discrimination reading attributes it to group-based inference; a compensating-differentials reading attributes part of it to omitted job attributes.

## 4 Unemployment: frictions, matching, and the natural rate

### 4.1 Types of unemployment

- **Frictional**: short-term unemployment from the time it takes to match workers to jobs (search, mobility, entry).
- **Structural**: a mismatch between the skills or locations of workers and the available vacancies.
- **Cyclical**: unemployment due to a deficiency of aggregate demand over the business cycle.

## 4.2 The Mortensen–Pissarides matching model

Search frictions mean unemployed workers  $U$  and vacancies  $V$  do not meet instantly. The flow of new hires is given by a matching function

$$M = m(U, V),$$

increasing in both arguments and typically taken to have constant returns to scale. Define labour-market tightness  $\theta = V/U$ . The job-finding rate is  $M/U = m(1, \theta)$  and the vacancy-filling rate is  $M/V = m(1/\theta, 1)$ . Equilibrium unemployment is where the flow into unemployment (separations) equals the flow out (matches): with separation rate  $\delta$  and labour force normalised to one,

$$\delta(1 - u) = m(u, V).$$

Vacancy creation is determined by free entry: firms post vacancies until the expected cost of an unfilled vacancy equals the expected profit from a filled one.

## 4.3 The Beveridge curve

The Beveridge curve is the empirical downward-sloping relation between the vacancy rate and the unemployment rate.

- Movements **along** the curve trace the business cycle: in a downturn vacancies fall and unemployment rises (lower-right); in a boom the reverse.
- An **outward shift** of the curve (higher unemployment at any vacancy rate) signals reduced matching efficiency, i.e. a rise in structural unemployment.

Reading a point: high vacancies together with high unemployment indicate a matching problem (structural); low vacancies with high unemployment indicate deficient demand (cyclical).

## 4.4 NAIRU

The NAIRU (non-accelerating-inflation rate of unemployment) is the equilibrium unemployment rate consistent with stable inflation — the labour-market counterpart of the natural rate. Frictional and structural unemployment persist at the NAIRU; cyclical unemployment is the deviation from it.

# 5 Institutional effects on equilibrium unemployment

## 5.1 Minimum wage, unions, and unemployment insurance

In the competitive model, a binding wage floor above the market-clearing wage creates unemployment by raising labour supply above labour demand:

- **Minimum wage:** a binding minimum reduces employment and creates a surplus of labour in the competitive model; its size depends on labour-demand elasticity.
- **Unions:** collective bargaining raises wages above the competitive level for members (the union wage premium), which can lower employment in the unionised sector and shift it to the non-union sector.
- **Unemployment insurance:** replacement income during search raises the reservation wage and lengthens search duration, raising frictional unemployment, while providing consumption insurance and potentially improving match quality.

Comparative international data show that countries with higher minimum wages, broader union coverage, and more generous unemployment insurance do not uniformly have higher unemployment, which motivates re-examining the simple competitive prediction.

## 5.2 The minimum-wage debate and the Card–Krueger reframing

The competitive model predicts that a higher minimum wage reduces employment. Card and Krueger studied a 1992 increase in New Jersey’s minimum wage using neighbouring Pennsylvania (which did not raise its minimum) as a comparison and found no significant employment loss in fast-food restaurants.

**Definition 4 (What the natural-experiment evidence changed)** The finding reframed the debate in two ways:

- empirically, modest minimum-wage increases were shown to have small or negligible employment effects, contradicting the simple competitive prediction;
- theoretically, it revived models with employer wage-setting power (monopsony), in which a minimum wage can raise both wages and employment over a range.

Methodologically, it helped establish the natural-experiment standard now expected in the field.

## 6 Inequality measurement

### 6.1 The Lorenz curve

Rank the population from poorest to richest. The Lorenz curve plots the cumulative share of total income  $L(p)$  held by the poorest fraction  $p$  of the population. It runs from  $(0, 0)$  to  $(1, 1)$ , lies on or below the  $45^\circ$  line of perfect equality, and is convex. The farther the curve lies below the diagonal, the greater the inequality.

### 6.2 The Gini coefficient

**Definition 5 (Gini coefficient)** The Gini coefficient is twice the area between the line of perfect equality and the Lorenz curve:

$$G = 2 \int_0^1 (p - L(p)) \, dp = 1 - 2 \int_0^1 L(p) \, dp.$$

It ranges from 0 (perfect equality) to 1 (perfect inequality). For discrete data with cumulative population shares  $p_0 = 0, p_1, \dots, p_n = 1$  and Lorenz ordinates  $L_0 = 0, L_1, \dots, L_n = 1$ , the trapezoidal approximation is

$$G \approx 1 - \sum_{i=1}^n (p_i - p_{i-1})(L_i + L_{i-1}).$$

**Interpretation since 1980.** Constructed on real distributional data, the Gini of disposable income has risen in many advanced economies since around 1980 (notably the United States and the United Kingdom), reflecting a widening of the income distribution; the trajectory differs across countries depending on labour-market and redistribution institutions.

## 7 Wealth inequality and its dynamics

### 7.1 Top concentration and the $r > g$ mechanism

Wealth is more concentrated than income: the top percentile holds a much larger share of wealth than of income. Piketty's central mechanism for rising wealth concentration is the comparison of the rate of return on capital  $r$  with the growth rate of the economy  $g$ .

**Proposition 6 (The  $r > g$  mechanism)** When the after-tax rate of return on wealth  $r$  exceeds the growth rate  $g$  of national income, accumulated and inherited wealth grows faster than incomes from labour. The capital/income ratio rises and wealth concentrates at the top, because past wealth is capitalised faster than new income is created. The steady-state capital/income ratio is  $\beta = s/g$  (saving rate over growth), which rises as growth slows.

### 7.2 Intergenerational mobility

Intergenerational mobility measures how strongly a child's economic position depends on the parents'. It is summarised by the intergenerational elasticity of earnings (the regression coefficient of child on parent log-earnings): a higher elasticity means lower mobility.

**Definition 7 (The Great Gatsby Curve)** Across countries, the intergenerational earnings elasticity is positively correlated with income inequality: more unequal countries tend to have lower intergenerational mobility. Higher inequality in one generation is associated with more persistent advantage across generations.

## 8 Causal inference for labour-market policy

### 8.1 The counterfactual problem

The causal effect of a policy on a unit is the difference between its outcome with the policy and its outcome without — but only one of these is ever observed. This is the fundamental problem of causal inference: the counterfactual is missing, and a simple before/after or treated/untreated comparison confounds the policy with everything else that differs.

### 8.2 Natural experiments and instrumental variables

A **natural experiment** uses variation in policy generated by an external event (legislation, a border, a lottery) that is plausibly unrelated to the outcome's other determinants, approximating random assignment.

An **instrumental variable**  $Z$  is a variable that affects the treatment  $D$  but influences the outcome  $Y$  only through  $D$  (relevance and exclusion). It identifies the causal effect by isolating the part of  $D$  driven by  $Z$ :

$$\text{effect} = \frac{\text{Cov}(Y, Z)}{\text{Cov}(D, Z)}.$$

### 8.3 Difference-in-differences

**Definition 8 (Difference-in-differences (DiD))** With a treated group and a control group observed before and after a policy, the DiD estimator is the difference of the two before–after changes:

$$\hat{\tau} = (\bar{Y}_{T,\text{post}} - \bar{Y}_{T,\text{pre}}) - (\bar{Y}_{C,\text{post}} - \bar{Y}_{C,\text{pre}}).$$

It removes fixed differences between groups and common time trends. **Identifying assumption — parallel trends:** absent the policy, treated and control groups would have followed the same trend. The Card–Krueger minimum-wage study is a difference-in-differences design.

**Common pitfall** A causal claim requires naming the identifying assumption and arguing it holds; correlation alone (a raw treated–control or before–after gap) does not identify the effect. DiD fails if the parallel-trends assumption is violated.

## 9 Automation and labour-market polarisation

### 9.1 Routine versus non-routine tasks

The task-based view classifies work by whether its tasks are **routine** (codifiable, rule-based) or **non-routine**, and whether they are **cognitive** or **manual**. Technology substitutes most readily for routine tasks, which can be programmed, and complements non-routine tasks.

### 9.2 Polarisation and complementarity

**Job polarisation** is the empirical hollowing-out of middle-skill, routine occupations relative to high-skill non-routine cognitive jobs and low-skill non-routine manual jobs. Where machines **complement** human labour (raising the productivity and demand for non-routine tasks), automation raises wages and employment in those tasks; where machines **substitute**, it displaces workers.

**Common pitfall** What economics can currently adjudicate: that routine tasks are most exposed to automation, and that recent decades show measurable polarisation. What remains genuinely uncertain: the net long-run effect on total employment and wages, because automation simultaneously displaces tasks and creates new ones, and the pace and reach of future technologies (including AI into non-routine cognitive tasks) are not known.